

Math 421: Homework 7

Solutions

Due: Oct 31, 2025

Problem 1: (Ross 11.4) Let

$$w_n = (-2)^n \text{ so that } (w_n) = (-2, 4, -8, 16, \dots)$$

$$x_n = 5^{(-1)^n} \text{ so that } (x_n) = (1/5, 5, 1/5, 5, \dots)$$

$$y_n = 1 + (-1)^n \text{ so that } (y_n) = (0, 2, 0, 2, \dots)$$

$$z_n = n \cos\left(\frac{n\pi}{4}\right) \text{ so that } (z_n) = (\sqrt{2}/2, 0, -3\sqrt{2}/2, -4, -5\sqrt{2}/2, 0, 7\sqrt{2}/2, 8, 9\sqrt{2}/2, \dots)$$

(a) Examples of monotone sequences.

w_n : Monotone sequence consists of even index terms $(w_{n_k}) = (4, 16, \dots)$.

x_n : Monotone sequence consists of even index terms $(x_{n_k}) = (5, 5, 5, \dots)$

y_n : Monotone sequence consists of even index terms $(y_{n_k}) = (2, 2, 2, \dots)$

z_n : Monotone sequence consists of indices of the form $4k - 2$, $(z_{n_k}) = (0, 0, 0, \dots)$

(b) Set of subsequential limits

$$w_n: \{-\infty, +\infty\}$$

$$x_n: \{1/5, 5\}$$

$$y_n: \{0, 2\}$$

$$z_n: \{-\infty, 0, +\infty\}$$

(c) $\limsup$ and $\liminf$

$$w_n: \liminf w_n = -\infty, \limsup w_n = +\infty$$

$$x_n: \liminf x_n = 1/5, \limsup x_n = 5$$

$$y_n: \liminf y_n = 0, \limsup y_n = 2$$

$$z_n: \liminf z_n = -\infty, \limsup z_n = +\infty$$

(d) None of the sequences converge or diverge to $\pm\infty$.

(e) The sequences x_n and y_n are bounded.

Problem 2: (Ross 11.8)

Theorem. For every sequence (s_n) , $\liminf s_n = -\limsup(-s_n)$.

Proof. Let (s_n) be a sequence and define $S_N = \{s_n : n > N\}$. From Exercise 5.4 we have that $\inf S_N = -\sup(-S_N)$, where $-S_N = \{-s_n : n > N\}$. By Definition 10.6 we have

$$\begin{aligned} \liminf s_n &= \lim_{N \rightarrow \infty} \inf\{s_n : n > N\} \\ &= \lim_{N \rightarrow \infty} (-\sup\{-s_n : n > N\}) \\ &= -\lim_{N \rightarrow \infty} \sup\{-s_n : n > N\} \\ &= -\limsup(-s_n) \end{aligned}$$

□

Problem 3: (Ross 11.10) Let (s_n) be defined as in Figure 11.12.

(a) $S = \{1, 1/2, 1/3, \dots, 0\}$

(b) $\limsup s_n = 1, \liminf s_n = 0$.

Problem 4: (Ross 12.2)

Theorem. For a sequence (s_n) , $\limsup |s_n| = 0$ if and only if $\lim s_n = 0$.

Proof. ($\Leftarrow$) Assume $\lim s_n = 0$, then for all $\epsilon > 0$ there exists N such that for all $n > N$ we have

$$|s_n - 0| = |s_n| < \epsilon$$

which in turn implies

$$||s_n| - 0| = |s_n| < \epsilon$$

so that $\lim |s_n| = 0$ and therefore, by Theorem 10.7, $\limsup |s_n| = 0$.

($\Rightarrow$) Assume $\limsup |s_n| = 0$. Since $|s_n| \geq 0$ for all n , we have $\liminf |s_n| \geq 0$ so that

$$0 \leq \liminf |s_n| \leq \limsup |s_n| = 0$$

which, by Theorem 10.7, implies $\lim |s_n| = 0$. Therefore, for all $\epsilon > 0$ there exists N such that for $n > N$ we have

$$||s_n| - 0| = |s_n| < \epsilon$$

which implies that $\lim s_n = 0$.

□

Problem 5: (Ross 12.3) Let (s_n) and (t_n) be the following sequences

$$(s_n) = (0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, \dots)$$

$$(t_n) = (2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, \dots)$$

(a) $\liminf s_n + \liminf t_n = 0 + 0 = 0$

(b) $\liminf(s_n + t_n) = 1$

(c) $\liminf s_n + \limsup t_n = 0 + 2 = 2$

(d) $\limsup(s_n + t_n) = 3$

(e) $\limsup s_n + \limsup t_n = 2 + 2 = 4$

(f) $\liminf(s_n t_n) = 0$

(g) $\limsup(s_n t_n) = 2$

Problem 6: (Ross 12.10)

Theorem. Prove (s_n) is bounded if and only if $\limsup |s_n| < +\infty$.

Proof. ($\Rightarrow$) If (s_n) is bounded, then there exists M such that $|s_n| \leq M$ for all n . Thus, for all N , M is an upper bound for the set $\{|s_n| : n > N\}$ so $\limsup |s_n| \leq \sup\{|s_n| : n > N\} \leq M$.

($\Leftarrow$) Let $\limsup |s_n| < +\infty$, then, since $|s_n| \geq 0$ for all n , we can take $0 \leq s = \limsup |s_n| < \infty$. Since

$$\limsup |s_n| = \limsup\{|s_n| : n > N\} = s$$

we have that for $\epsilon = 1$, there exists $N_0 \in \mathbb{N}$ such that for all $N > N_0$, we have

$$|\sup\{|s_n| : n > N\} - s| < 1$$

and

$$|\sup\{|s_n| : n > N\}| < 1 + |s|$$

Thus, for all $n > N > N_0$, we have

$$|s_n| < 1 + |s|.$$

Let $M = \max\{|s_1|, |s_2|, |s_3|, \dots, |s_{N_0}|, |s| + 1\}$, then for all n , $|s_n| \leq M$ and therefore (s_n) is bounded.

□